



DG INTERNS HUB

Week 3 - Advanced Cybersecurity Practical

Network Security Assessment Nmap, Wireshark, Vulnerability Assessment and Hardening

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Week	3
Lab	Kali Linux (10.0.0.2) vs Metasploitable2 (10.0.0.4)
Submission Date	21 September 2026

All testing was performed only on an isolated, authorized lab network that I own. No third-party or public systems were scanned.

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1. Introduction

Modern networks expose services that attackers can find in seconds. A security analyst must be able to discover devices, identify exposed services, read network traffic, judge risk, and prove that a fix worked. This report documents a small internal network assessment that follows exactly that loop: **discover, analyze, assess risk, harden, verify, document**.

The target is Metasploitable2, a deliberately vulnerable Linux machine. The analyst machine is Kali Linux. Both run as virtual machines on a private lab network (10.0.0.0/24). Every finding in this report comes from screenshots captured during the lab. Where evidence is missing or partial, the report says so instead of guessing.

2. Objectives

- Discover hosts and identify exposed services using Nmap (Tasks 7 and 8).
- Capture and analyze DNS, TCP, UDP, ICMP, ARP and HTTP traffic with Wireshark (Task 9).
- Match Nmap scan results with packet-level evidence (Task 10).
- Investigate traffic patterns and note anything that needs a closer look (Task 11).
- Build a risk-rated vulnerability assessment with at least 5 findings (Task 12).
- Apply at least 3 hardening changes and verify them with a re-scan (Task 13).
- Combine everything in a final mini security assessment (Task 14).

3. Lab Environment

Role	Hostname / OS	IP address	MAC address	Notes
Analyst	Kali Linux (VM)	10.0.0.2	08:00:27:5a:87:bc	Nmap 7.99, Wireshark 4.6.6, interface eth0
Target	Metasploitable2 (Linux 2.6.x, VM)	10.0.0.4	08:00:27:86:9c:2f	Hostnames seen: metasploitable.localdomain, irc.Metasploitable.LAN
Gateway	Virtual gateway / NAT	10.0.0.1	52:54:00:12:35:00	Seen in ARP and host discovery (QEMU NIC vendor)
DNS	External resolver	8.8.8.8	-	Used by Kali for name lookups

Network configuration: subnet 10.0.0.0/24, mask 255.255.255.0, broadcast 10.0.0.255. The target configuration was captured with *ifconfig* on Metasploitable2 (Figure 1). The Kali IP and the gateway are proven by Wireshark ARP packets and by the Nmap host discovery output. **Evidence gap:** a direct *ip a / ip route* screenshot from Kali was not captured.

```
eth0      Link encap:Ethernet  HWaddr 08:00:27:86:9c:2f
          inet addr:10.0.0.4  Bcast:10.0.0.255  Mask:255.255.255.0
          inet6 addr: fe80::a00:27ff:fe86:9c2f/64 Scope:Link
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:40 errors:0 dropped:0 overruns:0 frame:0
          TX packets:67 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:4957 (4.8 KB)  TX bytes:7196 (7.0 KB)
          Base address:0xd020 Memory:f0200000-f0220000

lo        Link encap:Local Loopback
          inet addr:127.0.0.1  Mask:255.0.0.0
          inet6 addr: ::1/128 Scope:Host
          UP LOOPBACK RUNNING  MTU:16436  Metric:1
          RX packets:97 errors:0 dropped:0 overruns:0 frame:0
          TX packets:97 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:0
          RX bytes:21529 (21.0 KB)  TX bytes:21529 (21.0 KB)

msfadmin@metasploitable:~$ sudo su
[sudo] password for msfadmin:
root@metasploitable:/home/msfadmin# cat/etc/inetd.conf| grep -E "ftptelnet|shell
lilogin|exec|ingreslock"
bash: cat/etc/inetd.conf: No such file or directory
root@metasploitable:/home/msfadmin#
```

Figure 1 - Target network configuration: eth0 10.0.0.4, mask 255.255.255.0, MAC 08:00:27:86:9c:2f

4. Network Diagram

Week 3 Lab Network Topology - 10.0.0.0/24

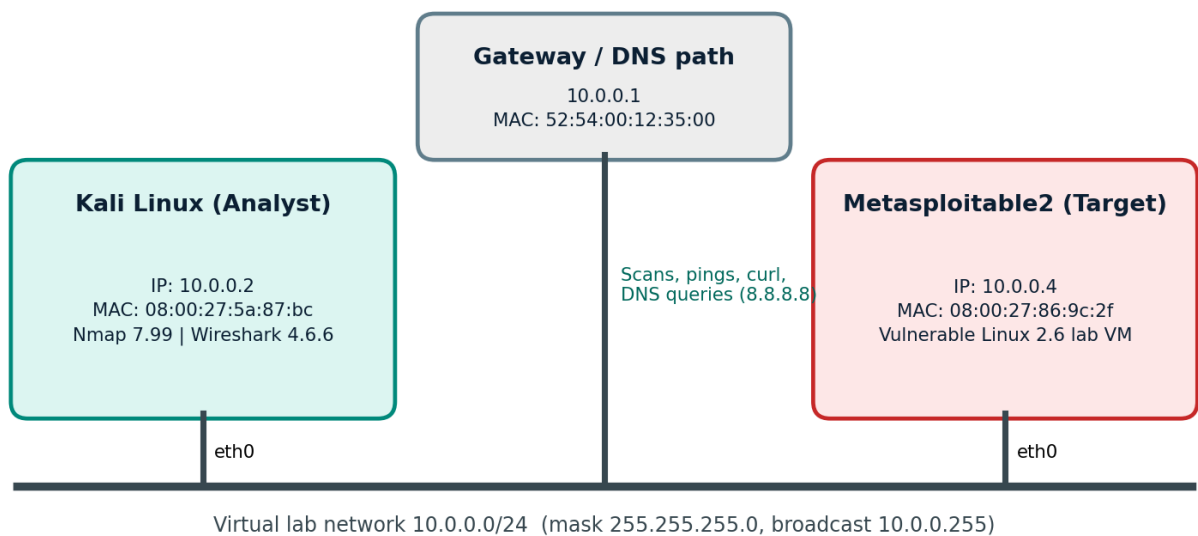


Figure 2 - Lab topology (addresses taken from ARP, ifconfig and Nmap output)

5. Tools Used

Tool	Version	Purpose in this lab
Nmap	7.99	Host discovery, port scanning, service/OS detection, NSE vulnerability scripts
Wireshark	4.6.6	Packet capture and display-filter analysis (dns, arp, icmp, tcp.flags.*)
Kali Linux	VM	Analyst workstation; ping, curl, nslookup used to create normal traffic
VirtualBox	-	Hosts both VMs (NIC vendor Oracle VirtualBox seen in scans)
Metasploitable2	VM	Authorized vulnerable target; inetd, xinetd, netstat, ps used for hardening
GitHub	-	Repository: cybersecurity-week-3-network-security

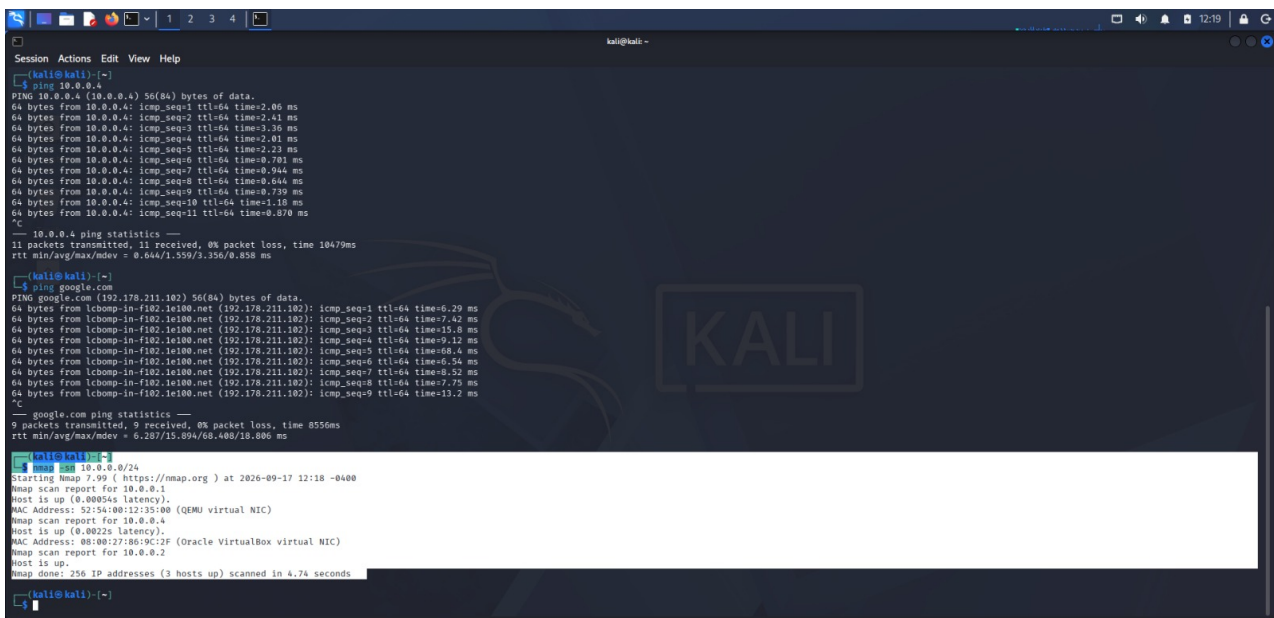
6. Nmap Methodology

- **Step 1 - Confirm connectivity:** ping the target (11 of 11 replies, 0% loss, 0.6-3.4 ms).
- **Step 2 - Host discovery:** `nmap -sn 10.0.0.0/24` to list live hosts.
- **Step 3 - Basic scan:** `nmap 10.0.0.4` (top 1000 TCP ports).
- **Step 4 - Advanced scans:** `-sV` versions, `-O` OS, `-sS SYN`, `-sU --top-ports 20 UDP`, `--script vuln NSE`.
- **Step 5 - Correlate** with Wireshark packets, then rate risk, harden and re-scan.
- Mapped to MITRE ATT&CK: T1595 Active Scanning (including T1595.001/.002), T1046 Network Service Discovery, T1018 Remote System Discovery.

7. Nmap Results (Task 7)

7.1 Host discovery

`nmap -sn 10.0.0.0/24` scanned 256 addresses and found **3 hosts up in 4.74 s**: 10.0.0.1 (MAC 52:54:00:12:35:00, QEMU virtual NIC), 10.0.0.4 (MAC 08:00:27:86:9c:2f, Oracle VirtualBox NIC) and 10.0.0.2 (the scanning machine itself).



```
kali@kali:~$ ping 10.0.0.4
PING 10.0.0.4 (10.0.0.4): 56(84) bytes of data:
64 bytes from 10.0.0.4: icmp_seq=1 ttl=64 time=2.06 ms
64 bytes from 10.0.0.4: icmp_seq=2 ttl=64 time=2.41 ms
64 bytes from 10.0.0.4: icmp_seq=3 ttl=64 time=2.36 ms
64 bytes from 10.0.0.4: icmp_seq=4 ttl=64 time=2.01 ms
64 bytes from 10.0.0.4: icmp_seq=5 ttl=64 time=2.23 ms
64 bytes from 10.0.0.4: icmp_seq=6 ttl=64 time=0.701 ms
64 bytes from 10.0.0.4: icmp_seq=7 ttl=64 time=0.944 ms
64 bytes from 10.0.0.4: icmp_seq=8 ttl=64 time=0.644 ms
64 bytes from 10.0.0.4: icmp_seq=9 ttl=64 time=0.799 ms
64 bytes from 10.0.0.4: icmp_seq=10 ttl=64 time=1.18 ms
64 bytes from 10.0.0.4: icmp_seq=11 ttl=64 time=0.878 ms
^C
--- 10.0.0.4 ping statistics ---
11 packets transmitted, 11 received, 0% packet loss, time 10479ms
rtt min/avg/max/mdev = 0.644/1.359/3.356/0.858 ms

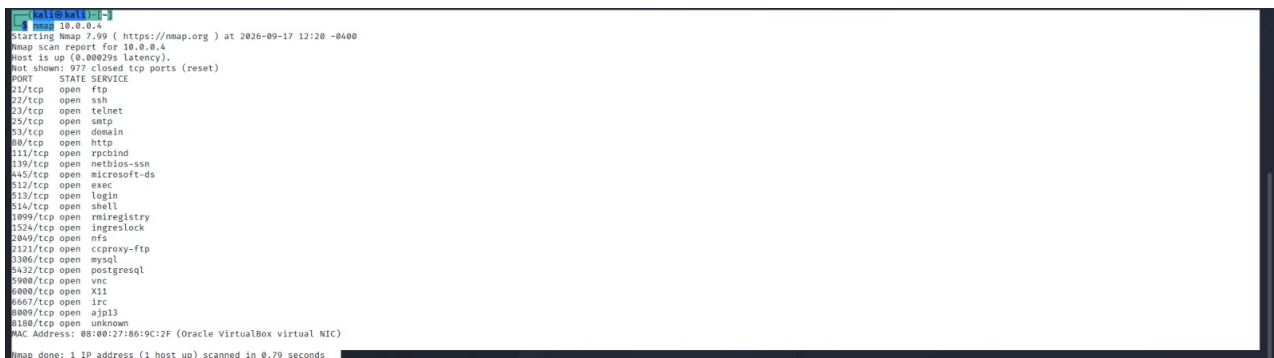
kali@kali:~$ ping google.com
PING google.com (192.178.211.102): 56(84) bytes of data:
64 bytes from lcbomp-in-f102.1e100.net (192.178.211.102): icmp_seq=1 ttl=64 time=6.29 ms
64 bytes from lcbomp-in-f102.1e100.net (192.178.211.102): icmp_seq=2 ttl=64 time=7.42 ms
64 bytes from lcbomp-in-f102.1e100.net (192.178.211.102): icmp_seq=3 ttl=64 time=15.8 ms
64 bytes from lcbomp-in-f102.1e100.net (192.178.211.102): icmp_seq=4 ttl=64 time=9.12 ms
64 bytes from lcbomp-in-f102.1e100.net (192.178.211.102): icmp_seq=5 ttl=64 time=68.4 ms
64 bytes from lcbomp-in-f102.1e100.net (192.178.211.102): icmp_seq=6 ttl=64 time=6.54 ms
64 bytes from lcbomp-in-f102.1e100.net (192.178.211.102): icmp_seq=7 ttl=64 time=8.52 ms
64 bytes from lcbomp-in-f102.1e100.net (192.178.211.102): icmp_seq=8 ttl=64 time=7.75 ms
64 bytes from lcbomp-in-f102.1e100.net (192.178.211.102): icmp_seq=9 ttl=64 time=13.2 ms
^C
--- google.com ping statistics ---
9 packets transmitted, 9 received, 0% packet loss, time 8556ms
rtt min/avg/max/mdev = 6.287/15.894/68.408/18.806 ms

kali@kali:~$ nmap 10.0.0.0/24
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-17 12:18 -0400
Nmap scan report for 10.0.0.1
Host is up (0.00054s latency).
MAC Address: 52:54:00:12:35:00 (QEMU virtual NIC)
Nmap scan report for 10.0.0.4
Host is up (0.0022s latency).
MAC Address: 08:00:27:86:9C:2F (Oracle VirtualBox virtual NIC)
Nmap scan report for 10.0.0.2
Host is up.
Nmap done: 256 IP addresses (3 hosts up) scanned in 4.74 seconds
```

Figure 3 - Ping tests and Nmap host discovery (3 hosts up)

7.2 Basic scan of 10.0.0.4

`nmap 10.0.0.4` reported **23 open TCP ports** and 977 closed ports (reset) in 0.79 s. This is a very large attack surface for one host.



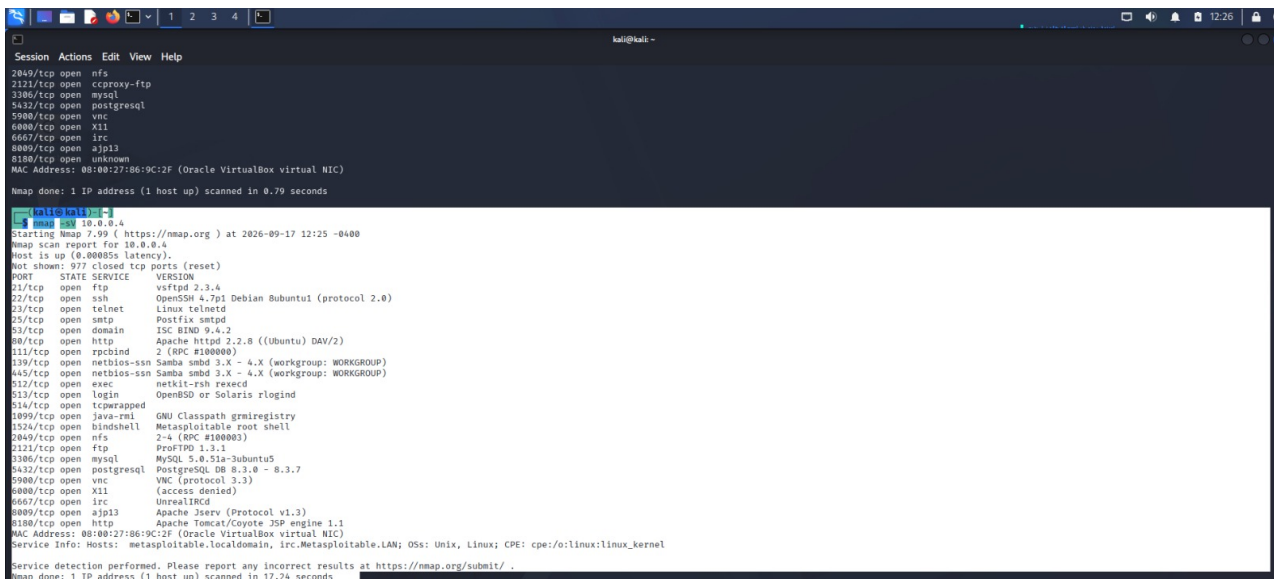
```
kali@kali:~$ nmap 10.0.0.4
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-17 12:20 -0400
Nmap scan report for 10.0.0.4
Host is up (0.00029s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
1099/tcp  open  rmiregistry
1524/tcp  open  ingreslock
2049/tcp  open  nfs
2121/tcp  open  cproxy-ftp
3306/tcp  open  mysql
5432/tcp  open  postgresql
5900/tcp  open  vnc
6000/tcp  open  X11
6667/tcp  open  irc
8009/tcp  open  ajp13
8100/tcp  open  unknown
MAC Address: 08:00:27:86:9C:2F (Oracle VirtualBox virtual NIC)
Nmap done: 1 IP address (1 host up) scanned in 0.79 seconds
```

Figure 4 - Basic Nmap scan: 23 open TCP ports

Port	Proto	Service	Port	Proto	Service
21	TCP	ftp	1099	TCP	rmiregistry
22	TCP	ssh	1524	TCP	ingreslock
23	TCP	telnet	2049	TCP	nfs
25	TCP	smtp	2121	TCP	ccproxy-ftp
53	TCP	domain	3306	TCP	mysql
80	TCP	http	5432	TCP	postgresql
111	TCP	rpcbind	5900	TCP	vnc
139	TCP	netbios-ssn	6000	TCP	X11
445	TCP	microsoft-ds	6667	TCP	irc
512	TCP	exec	8009	TCP	ajp13
513	TCP	login	8180	TCP	unknown
514	TCP	shell			

8. Advanced Nmap Analysis (Task 8)

8.1 Service and version detection (-sV)



```
Session Actions Edit View Help
2049/tcp open nfs
2121/tcp open ccrproxy-ftp
3306/tcp open mysql
5432/tcp open postgresql
5900/tcp open vnc
6000/tcp open x11
6667/tcp open irc
8009/tcp open ajp13
8180/tcp open unknown
MAC Address: 08:00:27:86:9C:2F (Oracle VirtualBox virtual NIC)
Nmap done: 1 IP address (1 host up) scanned in 0.79 seconds

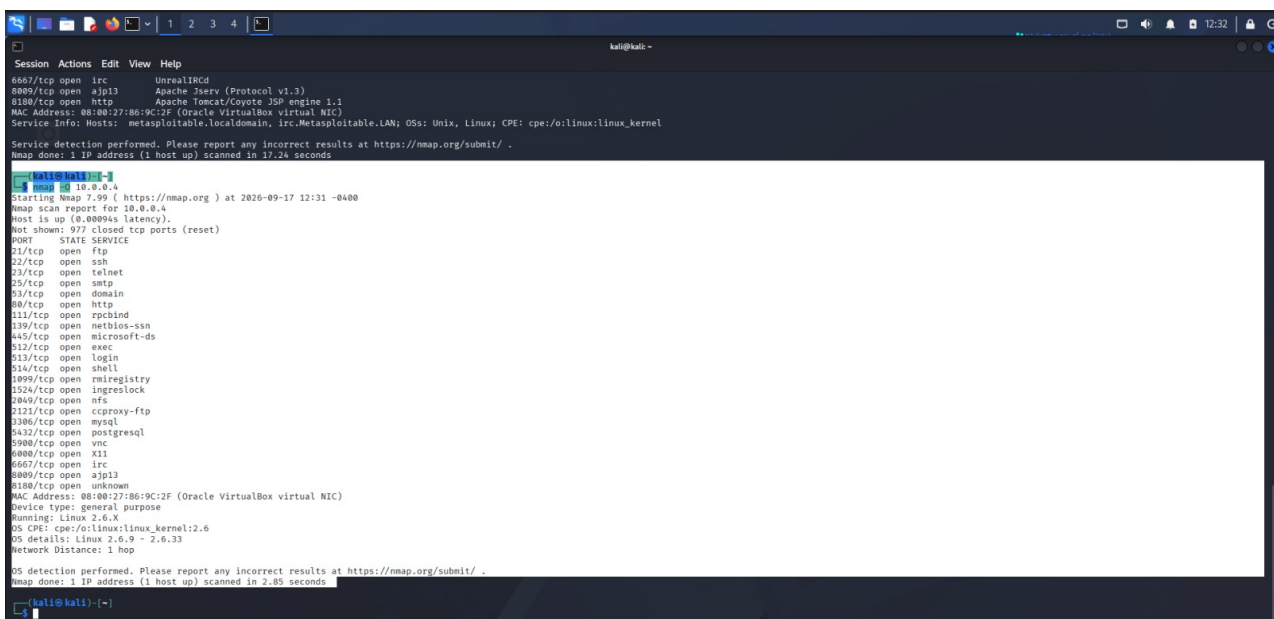
kali@kali:~$ nmap -sV 10.0.0.4
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-17 12:25 -0400
Nmap scan report for 10.0.0.4
Host is up (0.00005s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE        VERSION
21/tcp    open  ftp            vsftpd 2.3.4
22/tcp    open  ssh            OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
23/tcp    open  telnet         Linux telnetd
25/tcp    open  smtp           Postfix smtpd
53/tcp    open  domain         ISC BIND 9.4.2
80/tcp    open  http           Apache httpd 2.2.8 ((Ubuntu) DAV/2)
111/tcp   open  rpcbind        2 (RPC #100000)
139/tcp   open  netbios-ssn    Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp   open  netbios-ssn    Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
512/tcp   open  exec           netkit-rsh rexecd
513/tcp   open  login          OpenBSD or Solaris rlogind
514/tcp   open  tcpwrapped
1099/tcp  open  java-rmi       GNU Classpath gmicregistry
1524/tcp  open  bindshell      Metasploitable root shell
2049/tcp  open  nfs            2.4 (RPC #100003)
2121/tcp  open  ftp            ProFTPD 1.3.1
3306/tcp  open  mysql          MySQL 5.0.51a-Subuntus
5432/tcp  open  postgresql     PostgreSQL DB 8.3.0 - 8.3.7
5900/tcp  open  vnc            VNC (protocol 3.3)
6000/tcp  open  x11            (access denied)
6667/tcp  open  irc            UnrealIRCd
8009/tcp  open  ajp13          Apache Jserv (Protocol v1.3)
8180/tcp  open  http           Apache Tomcat/Coyote JSP engine 1.1
MAC Address: 08:00:27:86:9C:2F (Oracle VirtualBox virtual NIC)
Service Info: Hosts: metasploitable.localdomain, irc.Metasploitable.LAN; OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 17.24 seconds
```

Figure 5 - nmap -sV 10.0.0.4 (17.24 s)

8.2 OS detection (-O) and SYN scan (-sS)

nmap -O guessed **Linux 2.6.9 - 2.6.33** (CPE cpe:/o:linux:linux_kernel:2.6), device type general purpose, network distance 1 hop, scan time 2.85 s. A kernel from that era is more than 15 years out of support. The full **-sS** scan returned the same 23 open ports in 0.85 s.



```
Session Actions Edit View Help
6667/tcp open irc            UnrealIRCd
8009/tcp open ajp13         Apache Jserv (Protocol v1.3)
8180/tcp open http         Apache Tomcat/Coyote JSP engine 1.1
MAC Address: 08:00:27:86:9C:2F (Oracle VirtualBox virtual NIC)
Service Info: Hosts: metasploitable.localdomain, irc.Metasploitable.LAN; OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 17.24 seconds

kali@kali:~$ nmap -sS -O 10.0.0.4
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-17 12:31 -0400
Nmap scan report for 10.0.0.4
Host is up (0.00004s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE        VERSION
21/tcp    open  ftp            vsftpd 2.3.4
22/tcp    open  ssh            OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
23/tcp    open  telnet         Linux telnetd
25/tcp    open  smtp           Postfix smtpd
53/tcp    open  domain         ISC BIND 9.4.2
80/tcp    open  http           Apache httpd 2.2.8 ((Ubuntu) DAV/2)
111/tcp   open  rpcbind        2 (RPC #100000)
139/tcp   open  netbios-ssn    Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp   open  netbios-ssn    Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
512/tcp   open  exec           netkit-rsh rexecd
513/tcp   open  login          OpenBSD or Solaris rlogind
514/tcp   open  shell
1099/tcp  open  rmiregistry    GNU Classpath gmicregistry
1524/tcp  open  ingreslock
2049/tcp  open  nfs            2.4 (RPC #100003)
2121/tcp  open  ccrproxy-ftp
3306/tcp  open  mysql          MySQL 5.0.51a-Subuntus
5432/tcp  open  postgresql     PostgreSQL DB 8.3.0 - 8.3.7
5900/tcp  open  vnc            VNC (protocol 3.3)
6000/tcp  open  x11            (access denied)
6667/tcp  open  irc            UnrealIRCd
8009/tcp  open  ajp13          Apache Jserv (Protocol v1.3)
8180/tcp  open  http           Apache Tomcat/Coyote JSP engine 1.1
MAC Address: 08:00:27:86:9C:2F (Oracle VirtualBox virtual NIC)
Device type: general purpose
Running: Linux 2.6.x
OS CPE: cpe:/o:linux:linux_kernel:2.6
OS details: Linux 2.6.9 - 2.6.33
Network Distance: 1 hop

OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 2.85 seconds

kali@kali:~$
```

Figure 6 - OS detection

```
kali@kali:~$ nmap -sS 10.0.0.4
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-17 12:34 -0400
Nmap scan report for 10.0.0.4
Host is up (0.00081s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
1099/tcp  open  rmiregistry
1524/tcp  open  ingreslock
2049/tcp  open  nfs
2121/tcp  open  ccproxy-ftp
3306/tcp  open  mysql
5432/tcp  open  postgresql
5900/tcp  open  vnc
6000/tcp  open  x11
6667/tcp  open  irc
8009/tcp  open  ajp13
8180/tcp  open  unknown
MAC Address: 08:00:27:86:9C:2F (Oracle VirtualBox virtual NIC)
Nmap done: 1 IP address (1 host up) scanned in 0.98 seconds

kali@kali:~$ nmap -sS 10.0.0.4
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-17 12:34 -0400
Nmap scan report for 10.0.0.4
Host is up (0.00081s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
1099/tcp  open  rmiregistry
1524/tcp  open  ingreslock
2049/tcp  open  nfs
2121/tcp  open  ccproxy-ftp
3306/tcp  open  mysql
5432/tcp  open  postgresql
5900/tcp  open  vnc
6000/tcp  open  x11
6667/tcp  open  irc
8009/tcp  open  ajp13
8180/tcp  open  unknown
MAC Address: 08:00:27:86:9C:2F (Oracle VirtualBox virtual NIC)
Nmap done: 1 IP address (1 host up) scanned in 0.85 seconds
```

Figure 7 - TCP SYN scan (-sS), 23 open ports

8.3 Limited UDP scan (-sU --top-ports 20)

The UDP scan took 7.24 s. Only **53/udp (domain)** and **137/udp (netbios-ns)** were confirmed open. Ports 68, 69 (tftp), 138, 514 (syslog), 1434, 1900, 4500 and 49152 were *open|filtered*, which means Nmap could not tell whether they are open or blocked. The rest were closed.

```
kali@kali:~$ nmap -sU --top-ports 20 10.0.0.4
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-17 12:35 -0400
Nmap scan report for 10.0.0.4
Host is up (0.00093s latency).
PORT      STATE SERVICE
53/udp    open  domain
67/udp    closed dhcpcd
68/udp    open|filtered dhcpcd
69/udp    open|filtered tftp
123/udp   closed ntp
135/udp   closed msrpc
137/udp   open  netbios-ns
138/udp   open|filtered netbios-dgm
139/udp   closed netbios-ssn
161/udp   closed snmp
162/udp   closed snmptrap
445/udp   closed microsoft-ds
500/udp   closed isakmp
514/udp   open|filtered syslog
520/udp   closed route
631/udp   closed ipsec
1434/udp  open|filtered ms-sql-m
1900/udp  open|filtered upnp
4500/udp  open|filtered nat-t-ike
49152/udp open|filtered unknown
MAC Address: 08:00:27:86:9C:2F (Oracle VirtualBox virtual NIC)
Nmap done: 1 IP address (1 host up) scanned in 7.24 seconds

kali@kali:~$
```

Figure 8 - UDP scan of the top 20 ports

8.4 Service table (real results)

IP	Port	Proto	Service	Version (from -sV)	Risk	Observation
10.0.0.4	21	TCP	FTP	vsftpd 2.3.4	Critical	Version has a known backdoor (CVE-2011-2523)
10.0.0.4	22	TCP	SSH	OpenSSH 4.7p1 Debian 8ubuntu1	Medium	Needed for admin but very old version
10.0.0.4	23	TCP	Telnet	Linux telnetd	High	Clear-text remote login
10.0.0.4	25	TCP	SMTP	Postfix smtpd	Medium	SSLv3 POODLE reported by NSE
10.0.0.4	53	TCP	DNS	ISC BIND 9.4.2	Medium	Very old DNS server
10.0.0.4	80	TCP	HTTP	Apache httpd 2.2.8 (Ubuntu) DAV/2	High	Old web stack, vulnerable apps, default login shown
10.0.0.4	111 / 2049	TCP	rpcbind / NFS	RPC #100000 / 2-4 (RPC #100003)	Medium	Network file sharing exposed
10.0.0.4	139 / 445	TCP	SMB	Samba smbd 3.X - 4.X	Medium	File sharing exposed on legacy Samba
10.0.0.4	512-514	TCP	exec/login/shell	netkit-rsh rexecd / rlogind / tcpwrapped	High	Legacy r-services, no encryption
10.0.0.4	1099	TCP	Java RMI	GNU Classpath grmiregistry	High	NSE: default config allows remote code execution
10.0.0.4	1524	TCP	ingreslock	Metasploitable root shell	Critical	Direct root shell listening
10.0.0.4	2121	TCP	FTP	ProFTPD 1.3.1	Medium	Second FTP server
10.0.0.4	3306	TCP	MySQL	MySQL 5.0.51a-3ubuntu5	High	Database reachable from the whole LAN
10.0.0.4	5432	TCP	PostgreSQL	PostgreSQL DB 8.3.0 - 8.3.7	High	Database reachable from the whole LAN
10.0.0.4	5900 / 6000	TCP	VNC / X11	VNC (protocol 3.3) / access denied	Medium	Remote desktop / X display exposed
10.0.0.4	6667	TCP	IRC	UnrealIRCd	High	Chat daemon with a known backdoored release; version not confirmed
10.0.0.4	8009 / 8180	TCP	AJP / HTTP	Apache Jserv 1.3 / Tomcat-Coyote JSP 1.1	Medium	Tomcat manager page returns 401 (login page exists)
10.0.0.4	53 / 137	UDP	DNS / NetBIOS-NS	-	Low	Only two UDP ports confirmed open

8.5 Five questions for the important ports

Port / service	1. What is it?	2. Why needed?	3. Must it be exposed?	4. Impact if vulnerable	5. How to secure
21 FTP (vsftpd 2.3.4)	File transfer server	Old file sharing	No	Backdoor gives a root shell (NSE returned uid=0)	Remove; use SFTP over SSH; upgrade
22 SSH	Encrypted remote admin	Yes, for admins	Only from an admin network	Password guessing, old-version flaws	Update OpenSSH, key login, restrict source IPs
23 Telnet	Clear-text remote login	No, SSH replaces it	No	Passwords sniffable on the wire	Disable; use SSH
80 HTTP (Apache 2.2.8)	Web server with test apps	Lab web content	Only if needed	SQLi, XSS, file upload, default login msfadmin/msfadmin	Update, remove test apps, HTTPS, WAF
139/445 Samba	Windows file sharing	Not in this lab	No	Remote exploits, data leaks	Disable or firewall to trusted hosts
512-514 r-services	Old remote shell family	No	No	Trust abuse, clear-text	Disable in inetd
1099 Java RMI	Java remote objects	Not in this lab	No	Remote code execution (NSE: VULNERABLE)	Stop service, bind to localhost, patch
1524 ingreslock	Bind shell	Never	No	Instant root access for anyone	Remove the inetd line
3306/5432 databases	MySQL / PostgreSQL	Only for local apps	No, keep local	Data theft, weak or default logins	Bind to 127.0.0.1, firewall, upgrade
6667 UnrealIRCd	IRC chat server	No	No	Known release contains a backdoor	Stop service, remove, patch

9. Wireshark Methodology

Wireshark 4.6.6 captured on **eth0** of the Kali VM. Normal traffic was generated with: `ping 10.0.0.4`, `ping google.com`, `nslookup google.com`, and `curl http://10.0.0.4`. The capture was saved as **task9-capture.pcapng**. Display filters used: `dns`, `tcp.flags.syn == 1`, `icmp`, `arp`, and `tcp.flags.syn==1 || tcp.flags.reset==1` for the Nmap test.

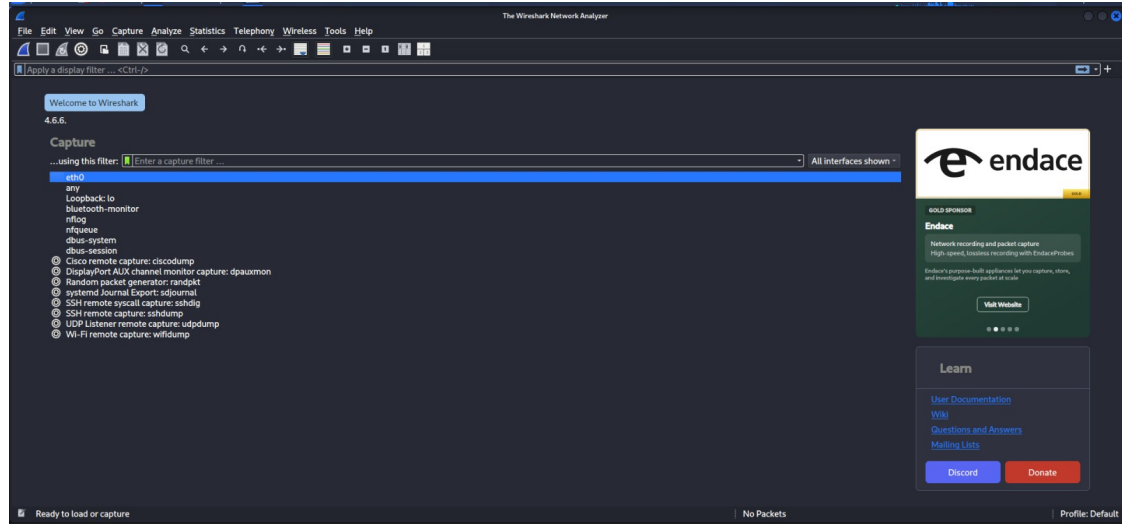


Figure 9 - Wireshark start page, eth0 selected

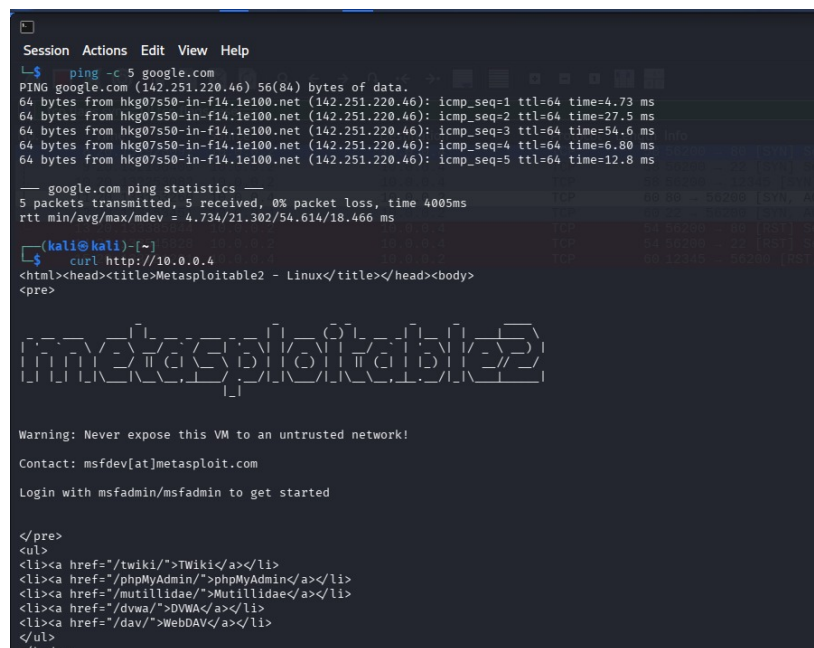
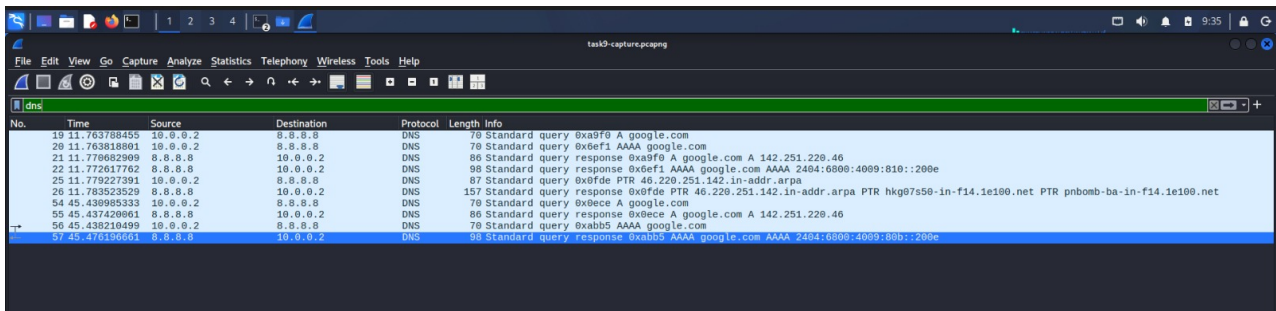


Figure 10 - Traffic generation: ping google.com and curl http://10.0.0.4 (shows Metasploit2 web page)

Observation: the web page fetched with curl is served in **clear text** and shows the default login `msfadmin/msfadmin` plus links to TWiki, phpMyAdmin, Mutillidae, DVWA and WebDAV. Anyone sniffing this traffic would read all of it.

10. Packet Analysis (Task 9)

10.1 DNS



The image shows a Wireshark packet capture window with the filter 'dns' applied. The packet list shows several DNS queries and responses. The packet details pane shows the structure of a DNS Standard query response.

No.	Time	Source	Destination	Protocol	Length	Info
19	11.763788455	10.0.0.2	8.8.8.8	DNS	70	Standard query 0xa9f0 A google.com
20	11.763818061	10.0.0.2	8.8.8.8	DNS	70	Standard query 0xbef1 AAAA google.com
21	11.77062909	8.8.8.8	10.0.0.2	DNS	86	Standard query response 0xa9f0 A google.com A 142.251.220.46
22	11.772617762	8.8.8.8	10.0.0.2	DNS	98	Standard query response 0xbef1 AAAA google.com AAAA 2404:6800:4009:80b::200e
25	11.779227391	10.0.0.2	8.8.8.8	DNS	87	Standard query 0xbfde PTR 46.220.251.142.in-addr.arpa
26	11.783523529	8.8.8.8	10.0.0.2	DNS	157	Standard query response 0xbfde PTR 46.220.251.142.in-addr.arpa PTR hkg07s50-in-f14.1e100.net PTR pnbomb-ba-in-f14.1e100.net
54	45.430985333	10.0.0.2	8.8.8.8	DNS	70	Standard query 0xece A google.com
55	45.437420661	8.8.8.8	10.0.0.2	DNS	86	Standard query response 0xece A google.com A 142.251.220.46
56	45.439210499	10.0.0.2	8.8.8.8	DNS	70	Standard query 0xab5 AAAA google.com
57	45.476195661	8.8.8.8	10.0.0.2	DNS	98	Standard query response 0xab5 AAAA google.com AAAA 2404:6800:4009:80b::200e

Figure 11 - Filter: dns

```
(kali@kali)-[~]
$ nslookup google.com
Server:      8.8.8.8
Address:     8.8.8.8#53

Non-authoritative answer:
Name:   google.com
Address: 142.251.220.46
Name:   google.com
Address: 2404:6800:4009:80b::200e
```

Figure 12 - nslookup google.com via 8.8.8.8 (A 142.251.220.46, AAAA 2404:6800:4009:80b::200e)

Kali sends every DNS query to **8.8.8.8**: A and AAAA for google.com, and a PTR lookup for 142.251.220.46 that resolves to hkg07s50-in-f14.1e100.net. Responses arrive within about 1 ms to 40 ms. DNS is plain UDP with no encryption, so anyone on the path can see which domains are requested.

10.2 ICMP

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000000	10.0.0.2	10.0.0.4	ICMP	98	Echo (ping) request id=0x2fca, seq=1/256, ttl=64 (reply in 2)
2	0.000000000	10.0.0.4	10.0.0.2	ICMP	98	Echo (ping) reply id=0x2fca, seq=1/256, ttl=64 (request in 1)
3	1.025789563	10.0.0.2	10.0.0.4	ICMP	98	Echo (ping) request id=0x2fca, seq=2/512, ttl=64 (reply in 4)
4	1.026523668	10.0.0.4	10.0.0.2	ICMP	98	Echo (ping) reply id=0x2fca, seq=2/512, ttl=64 (request in 3)
5	2.049557292	10.0.0.2	10.0.0.4	ICMP	98	Echo (ping) request id=0x2fca, seq=3/768, ttl=64 (reply in 6)
6	2.049981417	10.0.0.4	10.0.0.2	ICMP	98	Echo (ping) reply id=0x2fca, seq=3/768, ttl=64 (request in 5)
7	3.073640818	10.0.0.2	10.0.0.4	ICMP	98	Echo (ping) request id=0x2fca, seq=4/1024, ttl=64 (reply in 8)
8	3.074397796	10.0.0.4	10.0.0.2	ICMP	98	Echo (ping) reply id=0x2fca, seq=4/1024, ttl=64 (request in 7)
9	4.097722573	10.0.0.2	10.0.0.4	ICMP	98	Echo (ping) request id=0x2fca, seq=5/1280, ttl=64 (reply in 10)
10	4.098702596	10.0.0.4	10.0.0.2	ICMP	98	Echo (ping) reply id=0x2fca, seq=5/1280, ttl=64 (request in 9)
16	0.971004058	10.0.0.2	10.0.0.4	ICMP	370	Destination unreachable (Port unreachable)
23	11.774109670	10.0.0.2	142.251.220.46	ICMP	98	Echo (ping) request id=0x2fcb, seq=1/256, ttl=64 (reply in 24)
24	11.778833452	142.251.220.46	10.0.0.2	ICMP	98	Echo (ping) reply id=0x2fcb, seq=1/256, ttl=64 (request in 23)
27	12.775269493	10.0.0.2	142.251.220.46	ICMP	98	Echo (ping) request id=0x2fcb, seq=2/512, ttl=64 (reply in 28)
28	12.802779498	142.251.220.46	10.0.0.2	ICMP	98	Echo (ping) reply id=0x2fcb, seq=2/512, ttl=64 (request in 27)
29	13.777020964	10.0.0.2	142.251.220.46	ICMP	98	Echo (ping) request id=0x2fcb, seq=3/768, ttl=64 (reply in 30)
30	13.831621741	142.251.220.46	10.0.0.2	ICMP	98	Echo (ping) reply id=0x2fcb, seq=3/768, ttl=64 (request in 29)
31	14.779869546	10.0.0.2	142.251.220.46	ICMP	98	Echo (ping) request id=0x2fcb, seq=4/1024, ttl=64 (reply in 32)
32	14.780640990	142.251.220.46	10.0.0.2	ICMP	98	Echo (ping) reply id=0x2fcb, seq=4/1024, ttl=64 (request in 31)
34	14.970550371	10.0.0.2	10.0.0.4	ICMP	370	Destination unreachable (Port unreachable)
35	15.781460990	10.0.0.2	142.251.220.46	ICMP	98	Echo (ping) request id=0x2fcb, seq=5/1280, ttl=64 (reply in 36)
36	15.970227200	142.251.220.46	10.0.0.2	ICMP	98	Echo (ping) reply id=0x2fcb, seq=5/1280, ttl=64 (request in 35)
51	25.970510462	10.0.0.2	10.0.0.4	ICMP	370	Destination unreachable (Port unreachable)
53	36.978878943	10.0.0.2	10.0.0.4	ICMP	370	Destination unreachable (Port unreachable)
61	50.970538590	10.0.0.2	10.0.0.4	ICMP	370	Destination unreachable (Port unreachable)
65	61.970560241	10.0.0.2	10.0.0.4	ICMP	370	Destination unreachable (Port unreachable)
69	79.969915212	10.0.0.2	10.0.0.4	ICMP	370	Destination unreachable (Port unreachable)
77	95.97286253	10.0.0.2	10.0.0.4	ICMP	370	Destination unreachable (Port unreachable)
91	112.969286685	10.0.0.2	10.0.0.4	ICMP	370	Destination unreachable (Port unreachable)
101	124.974421580	10.0.0.2	10.0.0.4	ICMP	370	Destination unreachable (Port unreachable)
103	136.981009533	10.0.0.2	10.0.0.4	ICMP	370	Destination unreachable (Port unreachable)
105	144.985367973	10.0.0.2	10.0.0.4	ICMP	370	Destination unreachable (Port unreachable)

Figure 13 - Filter: icmp

Packets 1-10 are 5 echo request/reply pairs between 10.0.0.2 and 10.0.0.4. Packets 23-36 are echo pairs to 142.251.220.46 (google.com). Between them, many **Destination unreachable (Port unreachable)** messages go from 10.0.0.2 to 10.0.0.4 (see Task 11).

10.3 ARP

No.	Time	Source	Destination	Protocol	Length	Info
11	4.999552419	PCSSystemtec_86:9c:...	PCSSystemtec_5a:87:...	ARP	68	Who has 10.0.0.2? Tell 10.0.0.4
12	4.999578419	PCSSystemtec_5a:87:...	PCSSystemtec_86:9c:...	ARP	42	10.0.0.2 is at 08:00:27:5a:87:bc
13	5.057432977	PCSSystemtec_5a:87:...	PCSSystemtec_86:9c:...	ARP	42	Who has 10.0.0.4? Tell 10.0.0.2
14	5.058172813	PCSSystemtec_86:9c:...	PCSSystemtec_5a:87:...	ARP	68	10.0.0.4 is at 08:00:27:86:9c:2f
17	11.763497774	PCSSystemtec_5a:87:...	Broadcast	ARP	42	Who has 10.0.0.1? Tell 10.0.0.2
18	11.763733952	52:54:00:12:35:00	PCSSystemtec_5a:87:...	ARP	42	Who has 10.0.0.1? Tell 10.0.0.2
58	58.626423579	PCSSystemtec_5a:87:...	52:54:00:12:35:00	ARP	42	Who has 10.0.0.1? Tell 10.0.0.2
59	58.626683698	52:54:00:12:35:00	PCSSystemtec_5a:87:...	ARP	64	10.0.0.1 is at 52:54:00:12:35:00
62	55.9796803542	PCSSystemtec_86:9c:...	PCSSystemtec_5a:87:...	ARP	68	Who has 10.0.0.2? Tell 10.0.0.4
63	55.979619819	PCSSystemtec_5a:87:...	PCSSystemtec_86:9c:...	ARP	42	10.0.0.2 is at 08:00:27:5a:87:bc
66	67.010188318	PCSSystemtec_5a:87:...	PCSSystemtec_86:9c:...	ARP	42	Who has 10.0.0.4? Tell 10.0.0.2
67	67.011093963	PCSSystemtec_86:9c:...	PCSSystemtec_5a:87:...	ARP	68	10.0.0.4 is at 08:00:27:86:9c:2f
73	84.978981942	PCSSystemtec_86:9c:...	PCSSystemtec_5a:87:...	ARP	68	Who has 10.0.0.2? Tell 10.0.0.4
74	84.978921929	PCSSystemtec_5a:87:...	PCSSystemtec_86:9c:...	ARP	42	10.0.0.2 is at 08:00:27:5a:87:bc
96	117.971472677	PCSSystemtec_86:9c:...	PCSSystemtec_5a:87:...	ARP	68	Who has 10.0.0.2? Tell 10.0.0.4
97	117.971512439	PCSSystemtec_5a:87:...	PCSSystemtec_86:9c:...	ARP	42	10.0.0.2 is at 08:00:27:5a:87:bc
98	118.209648563	PCSSystemtec_5a:87:...	PCSSystemtec_86:9c:...	ARP	42	Who has 10.0.0.4? Tell 10.0.0.2
99	118.210315720	PCSSystemtec_86:9c:...	PCSSystemtec_5a:87:...	ARP	68	10.0.0.4 is at 08:00:27:86:9c:2f
106	149.986837615	PCSSystemtec_86:9c:...	PCSSystemtec_5a:87:...	ARP	68	Who has 10.0.0.2? Tell 10.0.0.4
107	149.986852037	PCSSystemtec_5a:87:...	PCSSystemtec_86:9c:...	ARP	42	10.0.0.2 is at 08:00:27:5a:87:bc

Figure 14 - Filter: arp

ARP resolves IP to MAC: 10.0.0.2 is at 08:00:27:5a:87:bc, 10.0.0.4 is at 08:00:27:86:9c:2f, gateway 10.0.0.1 is at 52:54:00:12:35:00. Requests repeat every few seconds to a minute as the ARP cache refreshes. ARP has no authentication, so on a real network it is a target for spoofing.

10.4 TCP and HTTP

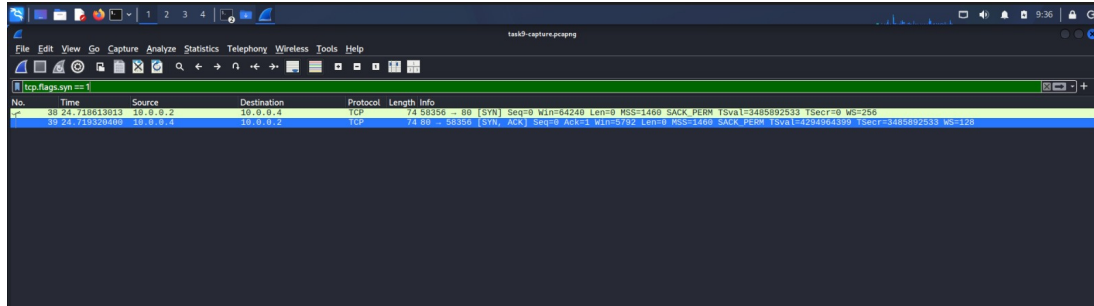


Figure 15 - Filter: tcp.flags.syn == 1

Packet 38 (24.718613 s): 10.0.0.2:58356 to 10.0.0.4:80 [SYN]. Packet 39: 10.0.0.4:80 to 10.0.0.2:58356 [SYN, ACK]. This is the first two steps of the TCP handshake for the curl request. It confirms port 80 accepts connections.

10.5 Key packet record

No.	Time (s)	Source	Destination	Proto	Src port	Dst port	Info	Security observation
19-22	11.76	10.0.0.2	8.8.8.8	DNS	not shown	53	A and AAAA query for google.com	Plain-text DNS to external resolver
25-26	11.78	10.0.0.2	8.8.8.8	DNS	not shown	53	PTR query for 142.251.220.46	Reverse lookup reveals host name
1-2	0.00	10.0.0.2	10.0.0.4	ICMP	-	-	Echo request/reply (id 0x2fca)	Target answers ping, so it is discoverable
11-14	4.99-5.06	10.0.0.2 / 10.0.0.4	each other	ARP	-	-	Who has / is at	Normal address resolution
38	24.72	10.0.0.2	10.0.0.4	TCP	58356	80	[SYN] Win=64240	Start of web request
39	24.72	10.0.0.4	10.0.0.2	TCP	80	58356	[SYN, ACK] Win=5792	Port 80 open

Protocol coverage: DNS, TCP, ICMP and ARP each have a filter screenshot. UDP is shown through DNS (which runs over UDP) and the ICMP port-unreachable messages. HTTP is proven by the curl output (Figure 10), but a dedicated *http* filter screenshot was not captured.

11. Nmap + Wireshark Investigation (Task 10)

While Wireshark captured on eth0, Kali ran a SYN scan against three ports: `sudo nmap -sS -p 22,80,12345 10.0.0.4` (Nmap 7.99, 19 Sep 2026, 0.61 s). Nmap reported 22/tcp open (ssh), 80/tcp open (http) and 12345/tcp closed (netbus).

```
(kali@kali)-[~]
└─$ sudo nmap -sS -p 22,80,12345 10.0.0.4
[sudo] password for kali:
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-19 09:40 -0400
Nmap scan report for 10.0.0.4
Host is up (0.0011s latency).

PORT      STATE SERVICE
22/tcp    open  ssh
80/tcp    open  http
12345/tcp  closed netbus
MAC Address: 08:00:27:86:9C:2F (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 0.61 seconds

(kali@kali)-[~]
```

Figure 16 - Nmap SYN scan output

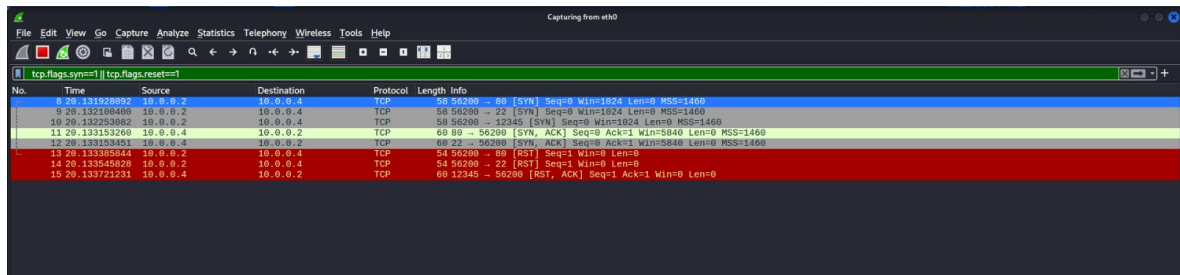


Figure 17 - Wireshark filter tcp.flags.syn==1 || tcp.flags.reset==1 during the scan

Pkt	Time (s)	Source:port	Destination:port	TCP flags	Meaning
8	20.131928	10.0.0.2:56200	10.0.0.4:80	SYN	Probe to port 80
9	20.132100	10.0.0.2:56200	10.0.0.4:22	SYN	Probe to port 22
10	20.132253	10.0.0.2:56200	10.0.0.4:12345	SYN	Probe to port 12345
11	20.133153	10.0.0.4:80	10.0.0.2:56200	SYN, ACK	Port 80 open
12	20.133451	10.0.0.4:22	10.0.0.2:56200	SYN, ACK	Port 22 open
13	20.133385	10.0.0.2:56200	10.0.0.4:80	RST	Kali resets: handshake not completed
14	20.133545	10.0.0.2:56200	10.0.0.4:22	RST	Kali resets: handshake not completed
15	20.133721	10.0.0.4:12345	10.0.0.2:56200	RST, ACK	Port 12345 closed

Comparison table (from real captured packets)

Nmap result	Wireshark evidence	Interpretation
Port 80 open	Pkt 8 SYN, then pkt 11 SYN/ACK, then pkt 13 RST	Service reachable; scanner did not finish the handshake (half-open scan)
Port 22 open	Pkt 9 SYN, then pkt 12 SYN/ACK, then pkt 14 RST	Service reachable; same half-open pattern
Port 12345 closed	Pkt 10 SYN, then pkt 15 RST/ACK from target	Port is not accepting connections
Port not responding (filtered)	Not captured in this lab	Would show a SYN with no reply, meaning a firewall drops packets. Recommended extra test

Explanation and conclusion

- **3 SYN packets, 3 answers, 2 resets:** 3 connection attempts, one per port. Every probe used the same source port **56200** and window size 1024, and all three were sent within about 0.3 ms. A person browsing never behaves like this.
- **Signature of Nmap -sS:** SYN, then SYN/ACK, then RST. A full connection would send ACK instead of RST, so the service never sees a real session and many applications do not log it. That is why it is called stealthy, but a firewall or IDS can still detect it by the burst of SYNs to many ports.
- **Conclusion:** Nmap results match the packets exactly. Open equals SYN/ACK. Closed equals RST/ACK. Wireshark is the ground truth that proves what the scanner really did.

12. Advanced Traffic Investigation (Task 11)

Source: my own capture **task9-capture.pcapng** (about 150 s of traffic visible in the screenshots) plus the Nmap capture from Task 10. No event below is labelled malicious. Each one is explained with evidence and marked as normal or worth a closer look.

Top communicating hosts

From the filtered views, all traffic is between **10.0.0.2 (Kali)**, **10.0.0.4 (target)**, the gateway 10.0.0.1, the DNS server 8.8.8.8 and 142.251.220.46 (google.com). Kali is the most active source (pings, DNS, curl, scan probes). **Evidence gap:** the Statistics, Conversations and Endpoints screenshots were not captured, so exact packet counts per host are not reported here.

Time (s)	Source IP	Destination IP	Port	Protocol	Observation	Security significance
6.97 to 144.99 (pkts 16, 34, 51, 53, 61, 65, 69, 77, 91, 101, 103, 105)	10.0.0.2	10.0.0.4	UDP (port not shown)	ICMP	12 Destination unreachable (Port unreachable) messages, every 8-18 s, length 370 bytes	Kali is rejecting UDP packets that the target keeps sending to a closed port. A regular repeating pattern can be normal (a periodic service) or a beacon. Needs investigation: expand the ICMP payload to read the original UDP port.
11.76-11.78 and 45.43-45.48	10.0.0.2	8.8.8.8	53	DNS/UDP	A, AAAA and PTR queries for google.com, all answered	Normal. Unencrypted DNS to an outside resolver; in production watch for long or random domain names (tunnelling).
11.77-15.79 (pkts 23-36)	10.0.0.2	142.251.220.46	-	ICMP	5 echo pairs to google.com, replies show TTL 64	Normal. A TTL of 64 from a distant host is unusual (far hosts usually show a lower value); the lab gateway or NAT may be answering. Worth noting.
4.99-5.06 and repeats to 149.98	10.0.0.2 / 10.0.0.4 / gateway	each other	-	ARP	Requests and replies repeat; each IP always maps to one MAC	Normal cache refresh. No duplicate IP or changing MAC seen, so no sign of ARP spoofing in this capture.
24.72 (pkts 38-39)	10.0.0.2	10.0.0.4	80	TCP/HTTP	SYN and SYN/ACK from curl; page shows default credentials	Plain HTTP exposes content and default login msfadmin/msfadmin to sniffers.
20.13 (pkts 8-15, Task 10 capture)	10.0.0.2	10.0.0.4	80, 22, 12345	TCP	SYN probes from one source port (56200), window 1024, then RST	Scan pattern: repeated attempts to several ports in under 2 ms. Expected here (my own scan), but this is what an attacker scan looks like.

Checks with no finding

- **TCP:** only two clean handshakes and the Nmap probes were seen; no unexplained repeated resets outside the scan.
- **Unexpected protocols / large traffic:** none seen in the screenshots. The capture is short (about 2.5 minutes).

13. Vulnerability Assessment (Task 12)

Findings come from Tasks 7 to 11. Severity is my judgement, based on (1) how easy it is to exploit, (2) what the attacker gets, and (3) whether the evidence is confirmed by a scanner or only suspected.

#	Finding	Evidence	Risk	Potential impact	Recommendation
1	vsftpd 2.3.4 backdoor on 21/tcp	-sV: vsftpd 2.3.4. NSE: VULNERABLE (Exploitable), CVE-2011-2523, command <i>id</i> returned uid=0(root)	Critical	Remote unauthenticated root shell. Confirmed by scanner, trivial to exploit	Remove vsftpd, use SFTP, upgrade
2	Root bind shell (ingreslock) on 1524/tcp	-sV: Metasploitable root shell. inetd.conf line: ingreslock stream tcp nowait root /bin/bash bash -i	Critical	Anyone who connects gets a root shell with no password	Delete the inetd line
3	UnrealIRCd on 6667/tcp	-sV: UnrealIRCd (exact version not shown)	High	Some UnrealIRCd releases contain a backdoor giving remote commands. Rated High, not Critical, because the version was not confirmed	Stop service, verify version, remove
4	Java RMI registry default configuration on 1099/tcp	NSE rmi-vuln-classloader: VULNERABLE, remote class loading can lead to code execution	High	Remote code execution, confirmed by scanner	Disable or bind to localhost, patch
5	Clear-text remote services: Telnet 23, rexec 512, rlogin 513, rsh 514	Nmap: 23, 512, 513, 514 open. inetd.conf lines active	High	Credentials and sessions can be read on the network and services trust remote hosts	Disable; use SSH
6	Vulnerable web apps and default credentials on 80/8180	curl page shows msfadmin/msfadmin, TWiki, phpMyAdmin, DVWA, Mutillidae, WebDAV. NSE http-enum found many /admin/ paths, Tomcat /manager/html (401), /webdav/. httponly flag not set	High	Account takeover, SQL injection, file upload; Apache 2.2.8 is outdated	Remove test apps, change defaults, HTTPS, update
7	Databases exposed to the LAN: MySQL 3306, PostgreSQL 5432	-sV: MySQL 5.0.51a, PostgreSQL 8.3.0 - 8.3.7	High	Data theft or modification if credentials are weak; versions are very old	Bind to localhost, firewall, upgrade
8	SSLv3 POODLE on SMTP 25/tcp	NSE ssl-poodle: VULNERABLE, CVE-2014-3566. ssl-dh-params also flagged an anonymous Diffie-Hellman weakness (output partly cut off in screenshot)	Medium	Man-in-the-middle can recover encrypted data; needs a network position, so not Critical	Disable SSLv3, force TLS 1.2+
9	Legacy SMB/Samba, NFS and rpcbind exposed (139, 445, 111, 2049, UDP 137)	-sV: Samba smbd 3.X - 4.X, nfs 2-4. NSE smb-vuln-ms10-061 and ms10-054 returned false	Medium	File-share exposure and information leaks. Specific exploits tested were negative, which lowers the rating	Disable or restrict to trusted hosts
10	Remote desktop / display and outdated core services: VNC 5900, X11 6000, OpenSSH 4.7p1, BIND 9.4.2	-sV output	Medium	VNC protocol 3.3 is weak; old SSH and DNS versions carry known flaws. X11 said access denied, which limits impact	Update, restrict access
11	TFTP on UDP 69 left enabled	UDP scan: 69/udp open filtered. inetd.conf: tftp line still active after hardening	Low	Unauthenticated file transfer if reachable. Rated Low because state is uncertain and it serves only /srv/tftp	Disable tftp in inetd or restrict

14. Risk Classification

Level	Count	Findings	How I decided
Critical	2	#1, #2	Remote root access with no login, confirmed by scanner output or by the service definition
High	5	#3, #4, #5, #6, #7	Likely to lead to compromise or data loss; either confirmed vulnerable or clear-text/default-credential exposure
Medium	3	#8, #9, #10	Real weakness but needs extra conditions (network position, specific exploit) or specific tests came back negative
Low	1	#11	Uncertain port state and limited impact

15. Security Hardening (Task 13)

All changes were made on the Metasploitable2 VM (authorized lab) as root. Six changes were applied, more than the required three.

#	Security issue	Action taken	Commands / evidence
H1	Legacy inetd services: telnet, shell, login, exec, ingreslock	Commented out in /etc/inetd.conf, restarted openbsd-inetd	sed -i '/^telnet/s/^/##/' /etc/inetd.conf (same for shell, login, exec, ingreslock); /etc/init.d/openbsd-inetd restart
H2	vsftpd 2.3.4 backdoor on 21	Disabled vsftpd in xinetd (disable = yes), restarted xinetd; netstat no longer shows 0.0.0.0:21	sed -i '/\sdisable=yes/' /etc/xinetd.d/vsftpd; /etc/init.d/xinetd restart
H3	Second FTP server ProFTPD on 2121	Stopped the service	/etc/init.d/proftpd stop
H4	UnrealIRCd on 6667	Killed process (PID 4663)	kill -9 4663 (temporary: not persistent)
H5	Java RMI registry on 1099	Killed process (PID 4655); netstat shows nothing on :1099	kill -9 4655 (temporary: not persistent)
H6	Samba on 139/445	Stopped nmbd and smb	/etc/init.d/samba stop

Evidence: before

```
TX packets:97 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:0
RX bytes:21529 (21.0 KB) TX bytes:21529 (21.0 KB)

msfadmin@metasploitable:~$ sudo su
[sudo] password for msfadmin:
root@metasploitable:/home/msfadmin# cat/etc/inetd.conf| grep -E "ftp!telnet!shell!login!exec!ingreslock"
bash: cat/etc/inetd.conf: No such file or directory
root@metasploitable:/home/msfadmin# cat /etc/inetd.conf| grep -E "ftp!telnet!shell!login!exec!ingreslock"
telnet          stream  tcp        nowait  telnetd  /usr/sbin/tcpd  /usr/sbin/in.tel
netd
#<off># ftp      stream  tcp        nowait  root     /usr/sbin/tcpd  /usr/sbi
n/in.ftpd
tftp            dgram   udp        wait    nobody   /usr/sbin/tcpd  /usr/sbin/in.tft
pd /srv/tftp
shell           stream  tcp        nowait  root     /usr/sbin/tcpd  /usr/sbin/in.rsh
d
login           stream  tcp        nowait  root     /usr/sbin/tcpd  /usr/sbin/in.rlo
gind
exec            stream  tcp        nowait  root     /usr/sbin/tcpd  /usr/sbin/in.rex
ecd
ingreslock      stream  tcp        nowait  root     /bin/bash bash -i
root@metasploitable:/home/msfadmin# _
```

Figure 18 - BEFORE: /etc/inetd.conf shows telnet, shell, login, exec and ingreslock enabled

Evidence: actions

```
ecd
#ngreslock stream tcp nowait root /bin/bash bash -i
root@metasploitable:/home/msfadmin# sed -i '/^telnet/s/^/#/' /etc/inetd.conf
root@metasploitable:/home/msfadmin# sed -i '/^shell/s/^/#/' /etc/inetd.conf
root@metasploitable:/home/msfadmin# sed -i '/^login/s/^/#/' /etc/inetd.conf
root@metasploitable:/home/msfadmin# sed -i '/^exec/s/^/#/' /etc/inetd.conf
root@metasploitable:/home/msfadmin# sed -i '/^ngreslock/s/^/#/' /etc/inetd.conf

root@metasploitable:/home/msfadmin# cat /etc/inetd.conf
#off># netbios-ssn      stream  tcp      nowait  root    /usr/sbin/tcpd  /usr/sbin/
n/smbd
#telnet                stream  tcp      nowait  telnetd /usr/sbin/tcpd  /usr/sbin/in.tel
netd
#off># ftp              stream  tcp      nowait  root    /usr/sbin/tcpd  /usr/sbin/
n/in.ftpd
#tftp                  dgram  udp      wait     nobody  /usr/sbin/tcpd  /usr/sbin/in.tft
pd /srv/tftp
#shell                 stream  tcp      nowait  root    /usr/sbin/tcpd  /usr/sbin/in.rsh
d
#login                 stream  tcp      nowait  root    /usr/sbin/tcpd  /usr/sbin/in.rlo
gind
#exec                  stream  tcp      nowait  root    /usr/sbin/tcpd  /usr/sbin/in.rex
ecd
#ngreslock stream tcp nowait root /bin/bash bash -i
root@metasploitable:/home/msfadmin# _
```

Figure 19 - inetd services commented out (sed) and file re-checked

```
ecd
#ngreslock stream tcp nowait root /bin/bash bash -i
root@metasploitable:/home/msfadmin# /etc/init.d/openbsd-inetd restart
root@metasploitable:/home/msfadmin# /etc/init.d/vsftpd stop
bash: /etc/init.d/vsftpd: No such file or directory
root@metasploitable:/home/msfadmin# ps aux | grep -i vsftpd
root    4959  0.0  0.0   3084  776 tty1    R+   10:10   0:00 grep -i vsftpd
root@metasploitable:/home/msfadmin# ps aux | grep -i unreal
root    4663  0.0  0.1   8540  2400 ?        S    09:22   0:00 /usr/bin/unreal
lrcd
root    4961  0.0  0.0   3084  776 tty1    R+   10:10   0:00 grep -i unreal
root@metasploitable:/home/msfadmin# kill -9 4663
root@metasploitable:/home/msfadmin# netstat -tulpn | grep :21
tcp        0      0  0.0.0.0:*                LISTEN
4522/xinetd
tcp6       0      0  :::2121                :::*                LISTEN
4561/proftpd: (acce
root@metasploitable:/home/msfadmin# ls /etc/xinetd.d/
chargen daytime discard echo time vsftpd
root@metasploitable:/home/msfadmin# sed -i '/^disable=yes/' /etc/xinetd.d/vsf
tpd
root@metasploitable:/home/msfadmin# /etc/init.d/xinetd restart
* Stopping internet superserver xinetd [ OK ]
* Starting internet superserver xinetd [ OK ]
root@metasploitable:/home/msfadmin#
```

Figure 20 - vsftpd disabled in xinetd, xinetd restarted, ProFTPD and RMI checks

```
4522/xinetd
tcp6       0      0  :::2121                :::*                LISTEN
4561/proftpd: (acce
root@metasploitable:/home/msfadmin# ls /etc/xinetd.d/
chargen daytime discard echo time vsftpd
root@metasploitable:/home/msfadmin# sed -i '/^disable=yes/' /etc/xinetd.d/vsf
tpd
root@metasploitable:/home/msfadmin# /etc/init.d/xinetd restart
* Stopping internet superserver xinetd [ OK ]
* Starting internet superserver xinetd [ OK ]
root@metasploitable:/home/msfadmin# netstat -tulpn | grep :21
tcp        0      0  :::2121                :::*                LISTEN
4561/proftpd: (acce
root@metasploitable:/home/msfadmin# /etc/init.d/proftpd stop
bash: /etc/init.d/proftpd: No such file or directory
root@metasploitable:/home/msfadmin# /etc/init.d/proftpd stop
* Stopping ftp server proftpd [ OK ]
root@metasploitable:/home/msfadmin# ps aux | grep -i rmiregistry
root    4655  0.0  1.2  74540 26564 ?        Sl   09:22   0:00 /usr/bin/rmireg
istry
root    5175  0.0  0.0   3084  780 tty1    R+   10:54   0:00 grep -i rmiregi
stry
root@metasploitable:/home/msfadmin# kill -9 4655
root@metasploitable:/home/msfadmin# netstat -tulpn | grep :1099
root@metasploitable:/home/msfadmin#
```

Figure 21 - ProFTPD stopped, rmiregistry killed, nothing left on :1099

```
4561/proftpd: (acce
root@metasploitable:/home/msfadmin# ls /etc/xinetd.d/
chargen daytime discard echo time vsftpd
root@metasploitable:/home/msfadmin# sed -i '/^disable=yes/' /etc/xinetd.d/vsf
tpd
root@metasploitable:/home/msfadmin# /etc/init.d/xinetd restart
* Stopping internet superserver xinetd [ OK ]
* Starting internet superserver xinetd [ OK ]
root@metasploitable:/home/msfadmin# netstat -tulpn | grep :21
tcp        0      0  :::2121                :::*                LISTEN
4561/proftpd: (acce
root@metasploitable:/home/msfadmin# /etc/init.d/proftpd stop
bash: /etc/init.d/proftpd: No such file or directory
root@metasploitable:/home/msfadmin# /etc/init.d/proftpd stop
* Stopping ftp server proftpd [ OK ]
root@metasploitable:/home/msfadmin# ps aux | grep -i rmiregistry
root    4655  0.0  1.2  74540 26564 ?        Sl   09:22   0:00 /usr/bin/rmireg
istry
root    5175  0.0  0.0   3084  780 tty1    R+   10:54   0:00 grep -i rmiregi
stry
root@metasploitable:/home/msfadmin# kill -9 4655
root@metasploitable:/home/msfadmin# netstat -tulpn | grep :1099
root@metasploitable:/home/msfadmin# /etc/init.d/samba stop
Stopping Samba daemons: nmbd smbd
root@metasploitable:/home/msfadmin#
```

Figure 22 - Samba (nmbd, smbd) stopped

16. Before / After Comparison

Verification scan after hardening: `nmap -sV -p 21,23,139,445,1099,1524,2121,6667 10.0.0.4` (19 Sep 2026, 11:07). All eight ports returned **closed**. The earlier targeted scan at 09:49 and the full scans showed the same ports open.

```
(kali@kali)-[~]
$ nmap -sV -p 21,23,1099,1524,6667 10.0.0.4
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-19 09:49 -0400
Nmap scan report for 10.0.0.4
Host is up (0.00093s latency).

```

PORT	STATE	SERVICE	VERSION
21/tcp	open	ftp	vsftpd 2.3.4
23/tcp	open	telnet	Linux telnetd
1099/tcp	open	java-rmi	GNU Classpath grmiregistry
1524/tcp	open	bindshell	Metasploitable root shell
6667/tcp	open	irc	UnrealIRCd

```
MAC Address: 08:00:27:86:9C:2F (Oracle VirtualBox virtual NIC)
Service Info: Host: irc.Metasploitable.LAN; OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel
Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 6.80 seconds
```

Figure 23 - BEFORE hardening (09:49): FTP, Telnet, RMI, root shell and IRC are open

```
(kali@kali)-[~]
$ nmap -sV -p 21,23,139,445,1099,1524,2121,6667 10.0.0.4
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-19 11:07 -0400
Nmap scan report for 10.0.0.4
Host is up (0.0013s latency).

```

PORT	STATE	SERVICE	VERSION
21/tcp	closed	ftp	
23/tcp	closed	telnet	
139/tcp	closed	netbios-ssn	
445/tcp	closed	microsoft-ds	
1099/tcp	closed	rmiregistry	
1524/tcp	closed	ingreslock	
2121/tcp	closed	ccproxy-ftp	
6667/tcp	closed	irc	

```
MAC Address: 08:00:27:86:9C:2F (Oracle VirtualBox virtual NIC)
Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 0.72 seconds
```

Figure 24 - AFTER hardening (11:07): all 8 scanned ports are closed

Security issue	Before	Action taken	After	Improvement
vsftpd backdoor (21)	21/tcp open ftp vsftpd 2.3.4	Disabled via xinetd	21/tcp closed	Critical backdoor removed
Root bind shell (1524)	1524/tcp open, Metasploitable root shell	Commented out in inetd.conf	1524/tcp closed	Critical root shell removed
Telnet (23)	23/tcp open, Linux telnetd	Commented out in inetd.conf	23/tcp closed	Clear-text login removed
ProFTPD (2121)	2121/tcp open ftp	Service stopped	2121/tcp closed	Second FTP server removed
Samba (139, 445)	Both open	Samba stopped	Both closed	SMB exposure removed
Java RMI (1099)	1099/tcp open java-rmi	Process killed	1099/tcp closed	RCE exposure removed (temporary)
UnrealIRCd (6667)	6667/tcp open UnrealIRCd	Process killed	6667/tcp closed	Backdoor risk removed (temporary)

Attack surface

Open TCP ports before: **23**. Confirmed closed after: **8** (21, 23, 139, 445, 1099, 1524, 2121, 6667), a reduction of about **35%**. Ports 512, 513 and 514 were also disabled in `inetd.conf`, but they were not part of the verification scan, so they are not counted. **A final full-range re-scan was not captured**; the remaining ports (for example 22, 25, 53, 80, 3306, 5432, 5900, 6000, 8009, 8180) are still expected to be open.

Honest limits of this hardening

- **Not persistent:** stopping ProFTPD/Samba with init scripts and killing UnrealIRCd and miregistry will not survive a reboot. Disabling boot start is still required.
- **Persistent:** the inetd.conf edits and the xinetd *disable* = yes setting stay after reboot.
- **Left as is:** the tftp line in inetd.conf is still active.

17. Final Mini Security Assessment (Task 14)

Area	Answer
Devices found	10.0.0.1 (gateway), 10.0.0.2 (Kali, analyst) and 10.0.0.4 (Metasploitable2, target)
Services exposed	23 open TCP ports on the target, including FTP, Telnet, SMB, databases, Java RMI, VNC and a root bind shell. UDP: DNS (53) and NetBIOS-NS (137) open
Protocols observed	DNS, ICMP, ARP, TCP (HTTP handshake and scan probes), UDP (through DNS and ICMP port-unreachable)
Communication patterns	Kali to target for ping, curl and scans; Kali to 8.8.8.8 for DNS; Kali to google.com for ping; repeating ARP; 12 recurring ICMP port-unreachable messages worth investigating
Major findings	11 findings: 2 Critical, 5 High, 3 Medium, 1 Low
Highest risk	vsftpd 2.3.4 backdoor (confirmed root by NSE) and the ingreslock root shell
Supporting evidence	Nmap -sV and --script vuln output, inetd.conf contents, packet captures
Hardening done	6 changes; 8 ports confirmed closed by re-scan
Attack surface	Reduced from 23 open TCP ports to at most 15 (about 35% fewer confirmed), with the biggest risks removed

18. Security Recommendations

- 1. Retire or replace the target.** Metasploitable2 runs 15-year-old software. In a real network, decommission or rebuild such hosts and patch everything supported.
- 2. Apply least exposure.** Remove services nobody needs (FTP, Telnet, r-services, Samba, NFS, VNC, RMI, IRC). Bind databases to 127.0.0.1.
- 3. Add a default-deny firewall.** Use iptables or ufw to allow only required ports (for example 22 from an admin subnet) and log the rest.
- 4. Encrypt and replace defaults.** Use SSH and SFTP instead of Telnet and FTP, HTTPS instead of HTTP, disable SSLv3, and change default credentials such as msfadmin/msfadmin.
- 5. Make hardening permanent and monitored.** Disable services at boot, re-scan after every change, and add an IDS or log alert for bursts of SYN packets like the one in Task 10.

19. Challenges Faced

- Nmap `--script vuln` took 325 s and some scripts failed (sslv2-drown and smb-vuln-regsvc-dos returned errors), so those checks are inconclusive.
- The first FTP fix did not work: stopping vsftpd with init.d failed because the service runs under xinetd, and port 21 stayed open. I found it with netstat and fixed it with `disable = yes` in `/etc/xinetd.d/vsftpd`.
- A command typed as `cat/etc/inetd.conf` (no space) failed and had to be repeated.
- The ProFTPD stop command also failed once (wrong init script name) before `/etc/init.d/proftpd` worked.
- Evidence gaps I am reporting openly:** no ip a / ip route screenshot, no Conversations/Endpoints screenshot, no dedicated HTTP or UDP filter screenshot, visible capture about 150 s (task asks for 5-10 min), no filtered-port example, and no full-range scan after hardening.

20. Conclusion

The lab shows how quickly a security analyst can turn a network scan into a risk-ranked action plan. Nmap found 23 exposed TCP ports and two Critical flaws, including a backdoor that returned root. Wireshark proved what Nmap did on the wire: SYN/ACK for open ports, RST for closed ports, and a fast burst from one source port. Six hardening steps closed eight ports, and the re-scan confirmed it. The work also showed why verification matters, because the first FTP fix looked correct but port 21 was still open until netstat exposed it. The next step is a full re-scan, persistent service removal and a firewall.

21. References

- Nmap: <https://nmap.org>
- Nmap Reference Guide: <https://nmap.org/book/man.html>
- Nmap NSE: <https://nmap.org/book/nse.html>
- Wireshark: <https://www.wireshark.org>
- Wireshark User's Guide: https://www.wireshark.org/docs/wsug_html_chunked/
- Wireshark Display Filter Reference: <https://www.wireshark.org/docs/dfref/>
- MITRE ATT&CK: <https://attack.mitre.org>
- CVE Program: <https://www.cve.org>
- CVE-2011-2523 (vsftpd 2.3.4 backdoor), CVE-2014-3566 (POODLE)

22. Screenshots / Evidence Index

Task	Evidence	File
7	Network config (target)	01-Nmap/Screenshots/01_network_config_metasploitable_ifconfig.jpg
7	Ping + host discovery	01-Nmap/Screenshots/02_ping_and_host_discovery_nmap_sn.jpg
7	Basic scan	01-Nmap/Screenshots/03_nmap_basic_scan.jpg
8	Service versions	01-Nmap/Screenshots/04_nmap_sV_service_versions.jpg
8	OS detection	01-Nmap/Screenshots/05_nmap_O_os_detection.jpg
8	SYN scan	01-Nmap/Screenshots/06_nmap_sS_syn_scan.jpg
8	UDP scan	01-Nmap/Screenshots/07_nmap_sU_udp_top20.jpg
8/12	NSE vuln scan (3 parts)	01-Nmap/Screenshots/08_..., 09_..., 10_...
9	DNS, ICMP, ARP, TCP filters	02-Wireshark/Screenshots/02_ to 06_
10	Nmap -sS packets	02-Wireshark/Screenshots/08_ and 01-Nmap/.../11_
13	Before evidence	04-Hardening/Before/
13	Hardening actions and after scan	04-Hardening/After/

All screenshots are embedded in this report and stored in the project folders. No passwords, keys or personal data were uploaded. The only credentials visible are the public default Metasploitable2 login shown on its own web page.